Úloha 10

Zadání

Spočtěte $\mathrm{d}^3 f$, je-li $f(x, y, z) = xyz$.

Řešení

Nechť:

$$x_1 \coloneqq x \qquad x_2 \coloneqq y \qquad x_3 \coloneqq z$$

$$\begin{aligned}
\mathrm{d}^3 f = \sum_{i,j,k=1}^{3} \frac{\partial^3 x_1 x_2 x_3}{\partial x_i \partial x_j \partial x_k} \,\mathrm{d}x_i \,\mathrm{d}x_j \,\mathrm{d}x_k &= 0 + 0 + 0 + 0 + 0 + \,\mathrm{d}x_1 \,\mathrm{d}x_2 \,\mathrm{d}x_3 + \\
&+ 0 + \,\mathrm{d}x_1 \,\mathrm{d}x_3 \,\mathrm{d}x_2 + 0 + 0 + 0 + \,\mathrm{d}x_2 \,\mathrm{d}x_1 \,\mathrm{d}x_3 + 0 + 0 + 0 + \\
&+ \,\mathrm{d}x_2 \,\mathrm{d}x_3 \,\mathrm{d}x_1 + 0 + 0 + 0 + \,\mathrm{d}x_3 \,\mathrm{d}x_1 \,\mathrm{d}x_2 + 0 + \,\mathrm{d}x_3 \,\mathrm{d}x_2 \,\mathrm{d}x_1 + \\
&+ 0 + 0 + 0 + 0 + 0 \\
&= 6 \,\mathrm{d}x_1 \,\mathrm{d}x_2 \,\mathrm{d}x_3 = 6 \,\mathrm{d}x \,\mathrm{d}y \,\mathrm{d}z
\end{aligned}$$

Řešení

$$\mathrm{d}^3 f(x, y, z) = 6 \,\mathrm{d}x \,\mathrm{d}y \,\mathrm{d}z$$

Úloha 16

Zadání

Nalezněte obecné řešení rovnice:

$$(xy^2+y)\mathrm{d}x - x\mathrm{d}y = 0, \mu = \mu(y)$$

Řešení

$$\frac{\partial}{\partial y}(xy^2+y) = 2xy+1 \neq \frac{\partial}{\partial x}(-x) = -1$$

$$\begin{aligned}\frac{\mu'(\Psi)}{\mu(\Psi)} &= \frac{\frac{\partial}{\partial x}(-x) - \frac{\partial}{\partial y}(xy^2+y)}{(xy^2+y)\frac{\partial\Psi}{\partial y} - (-x)\frac{\partial\Psi}{\partial x}} \\ &= \frac{-2xy-2}{(xy^2+y)\frac{\partial\Psi}{\partial y} - (-x)\frac{\partial\Psi}{\partial x}}\end{aligned}$$

Nechť:

$$\Psi(y) = \ln y,$$

pak:

$$\frac{\mu'(\Psi)}{\mu(\Psi)} = \frac{-2xy-2}{xy+1} = -2$$

$$\begin{aligned}\ln(\mu(\Psi)) &= -2\Psi \\ \mu(\Psi) &= e^{-2\Psi} \\ \mu(y) &= e^{-2\ln y} \\ \mu(y) &= \frac{1}{y^2}\end{aligned}$$

$$\left(x+\frac{1}{y}\right)\mathrm{d}x - \frac{x}{y^2}\mathrm{d}y = 0$$

$$U(x,y) = \int \left(x + \frac{1}{y}\right)\,\mathrm{d}x = \frac{1}{2}x^2 + \frac{x}{y} + \phi(y)$$

$$-\frac{x}{y^2} = \frac{\partial U(x,y)}{\partial y} = -\frac{x}{y^2} + \phi'(y)$$

$$\phi'(y) = 0 \implies \phi(y) = \text{konst.}$$

$$U(x,y) = \frac{1}{2}x^2 + \frac{x}{y} + C = 0$$

$$y = \frac{2x}{C - x^2}$$

Výsledek

$$y = \frac{2x}{C - x^2}$$